Devesh Dhole
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LinkedIn

Education

- **Ph.D. Solar Astrophysics**, Northumbria University, Newcastle Upon Tyne *March 2025 - Current*
- **M.Sc. Physics**, Indian Institute of Technology Bombay, Mumbai *August 2022 - August 2024*
  CGPA: 8.47/10
- **B.E.(Hons.) Computer Science**, Birla Institute of Technology and Science, Goa *August 2015 - August 2019*
  CGPA: 7.03/10

Research Interests

- Modeling high-energy astrophysical systems, including accretion disks, using GRMHD and MHD simulations.
- Applying machine learning and AI techniques to astrophysical problems spanning image classification, numerical simulations, and gravitational wave analysis.
- Probing modified theories of gravity through analysis of quasi-normal mode (QNM) frequencies from compact binary mergers.
- Investigating turbulence in collisionless plasmas, such as the solar wind, through particle-in-cell (PIC) simulations.

Research Experience

Role of Current Sheets in Decaying Turbulence *March 2025 – Present*
PhD Student under Prof. Luca Franci *Northumbria University, Newcastle*

- Running large-scale hybrid PIC simulations (**CAMELIA**; kinetic ions, fluid electrons) on **Dirac HPC** to investigate the role of current sheets in decaying solar wind turbulence, linking their properties to reconnection, plasma heating, intermittency, and the ion-scale spectral break.

Modelling Accretion Disks in SpECTRE Code *August 2024 – Present*
Long-term Visiting Student under Prof. Prayush Kumar *TIFR-ICTS*

- Developing **GRMHD** simulations of magnetized **Fishbone-Moncrief** accretion disk using the **SpECTRE** code, and benchmarking physical consistency and stability by cross-validating with BHAC results.
- Leveraging SpECTRE's hybrid Finite Difference–**Discontinuous Galerkin** framework to improve simulation performance, accuracy, and scalability for modeling relativistic astrophysical flows.

Exploring Modified Gravity Theories through Neural Networks *August 2023 – August 2024*
Master's Thesis under Prof. Shankaranarayanan S *IIT Bombay*

- Employed **Physics-Informed Neural Networks (PINNs)** using **NeuroDiffEq** to solve Einstein's equations in modified gravity theories, with a focus on implications for **gravitational wave observations**.
- Investigated spherically symmetric vacuum solutions, revealing potential violations of classical results such as **Birkhoff's theorem** and the **no-hair theorem** in non-GR frameworks.
- Discovered a **non-singular Schwarzschild-like black hole solution** within **Stelle gravity**; ongoing work aims to understand its observational signatures in gravitational wave data.

Simulating AGN jets using PLUTO Code *June 2021 - August 2021*
Project under Prof. Bhargav Vaidya *IIT Indore*

- Performed **MHD** simulations of extragalactic jets using **PLUTO** code with **MPI** on **HPC** cluster to constrain simulation parameters using observations and visualized the results using **Visit** and **Paraview**.
- Injected Lagrangian particles into the simulation box using the particle module of the PLUTO code to compute non-thermal Synchrotron Emissivity at different viewing angles used to constrain the simulation parameters.

Coronal Hole Detection using Neural Networks *May 2023 - July 2023*
Project under Dr. Dipankar Banerjee(Director, ARIES) *ARIES, Nainital*

- Implemented the **U-net** architecture, a convolutional neural network (CNN), for precisely detecting coronal holes by leveraging data derived from **SDO/AIA 193 Å** UV images and **HMI** Magnetogram data.
- Assessed the model's performance through the application of the **Intersection over Union (IOU)** metric and **Binary Cross-Entropy** functions, ensuring a comprehensive evaluation of its effectiveness.

Feature Selection, Information Retrieval *August 2017 - December 2017*
Course Project under Prof. Rajendra Roul *BITS Goa*

- Devised an innovative feature selection approach by combining two novel methods, including dependent feature removal based on various correlation factors and an SVM-driven technique.
- Leveraged an approximation of **Probabilistic Neural Networks** for precise feature selection, consistently achieving an impressive average accuracy of approximately **85%** across diverse classifiers.

Key Projects

Recent Discoveries in Magnetar Physics, Advanced Astrophysics *October 2023 - December 2023*
Course Project under Prof. Archana Pai *IIT Bombay*

- Studied the various magnetar phenomena, including a spectrum of X-ray activities, short and long bursts, giant flares, quasi-periodic oscillations, glitches, and anti-glitches.
- Studied the current theoretical understanding of such diverse phenomena via observations and numerical simulations.

Gravitational Simulations using Python *August 2022 - November 2022*
Project under Krittika Astronomy Club *IIT Bombay*

- Applied the intricate theory of the three-body problem, meticulously deriving **Lagrange points**, **Zero Velocity Curves**, and **Poincaré maps** for profound insights into celestial mechanics.
- Used the equations of motion to simulate the paths of a particle perturbed slightly from the Lagrange points using the **Euler Richardson Method** and verified that they are constrained by the zero velocity curves.

Work Experience

Software Development Engineer, Flipkart, Bengaluru *July 2019 - September 2021*

- Led collaborative efforts across the system development life-cycle, orchestrating seamless transitions from requirements gathering to successful production releases and post-launch optimizations.
- Designed, implemented and maintained a scalable system architecture for high-availability web applications, managing server and database administration for optimal performance under threefold loads.

Software Development Intern, IBM, Bengaluru *July 2018 - December 2018*

- Developed a Kafka service proficient in publishing messages to a Kafka cluster and consuming them based on subscription, eliminating bottlenecks in traditional queuing systems like RabbitMQ and JMS.
- Significantly enhanced operational efficiency, achieving a performance boost of at least fivefold.

Workshops, Conferences and Summer Schools

- **National Astronomy Meeting, University of Birmingham, UK** *July 2026*
  Gave a **contributed talk**, *Current Sheet Properties and Their Dependence on the Plasma Conditions in Hybrid Turbulence Simulations*, presenting results from large-scale hybrid PIC simulations on how current sheet statistics depend on ion and electron plasma beta.
- **Postgraduate Research Conference, Northumbria University, Newcastle, UK** *June 2026*
  Presented a **poster** on current sheet detection methods and their dependence on plasma conditions in hybrid PIC simulations of decaying turbulence.
- **DiRAC Cross-Community AMD GPU Hackathon, University of Birmingham, UK** *March 2026*
  Ported key components of the hybrid PIC code **CAMELIA** to AMD GPUs in a three-day hands-on workshop, working alongside experts from AMD, DiRAC, UCL, and the University of Cambridge to accelerate performance using GPU technologies.
- **Autumn MIST 2025, Geological Society of London, UK** *November 2025*
  Participated in the UK's Magnetosphere, Ionosphere and Solar-Terrestrial community meeting, engaging with talks and posters spanning solar-wind turbulence, space plasmas, and magnetospheric dynamics.
- **Arcetri Workshop on Plasma Astrophysics, University of Florence, Italy** *October 2025*
  Engaged in discussions on solar-wind turbulence, kinetic-scale dissipation, reconnection, and plasma astrophysics in various contexts with an international research community
- **Transients, IIT-Bombay** *April 2024*
  A two-day meeting centred on science, hardware and software for high energy transients.
- **Gravitational-Wave Astronomy, ICTS-TIFR, Bangalore** *Aug 2023*
  Series of lectures on Numerical Relativistic Hydrodynamics and its applications in Gravitational Wave physics with hands-on sessions on simulations.
- **Computational MHD with PLUTO code, IISc Bangalore** *July 2023*
  Series of lectures on basics of fluids and computational methods along with hands-on simulation session with PLUTO.

Key Courses Undertaken

Physics	**Advanced Astrophysics, Astronomy and Astrophysics, General Theory of Relativity, Advanced Simulation Techniques in Physics, Non-Linear Dynamics**, Statistical Physics, Quantum Mechanics I & II, Electromagnetic Theory I & II, Nuclear & Particle Physics, Atomic and Molecular Physics, Condensed Matter Physics
Computer Science	**Computer Architecture**, **Computer Networks, Parallel Computing**, **Data Mining**, Database Systems, Theory of Computation, Operating Systems, Design and Analysis of Algorithms

Technical Skills

Programming	C++, Fortran, Bash, Python, Java, Latex, IDL, HTML, CSS
Software & IDEs	Mathematica, PLUTO, Visit, Paraview, Gnuplot, DS9, SpECTRE, Camelia
Technologies	NeuroDiffEq, TensorFlow, SunPy, PyTorch, Keras, MPI, OpenMP, MySQL, Git, Hadoop

Scholastic Achievements

- Secured an **All India Rank 23** out of 12740 candidates in IIT-JAM Physics *2022*
- Secured an **All India Rank 51** out of 4005 candidates in JEST Physics *2022*

Extracurriculars

- Conquered remarkable Himalayan altitudes, including Marsar Ridge at 13,500 ft, Hampta Pass at 14,065 ft, and the formidable Mount Friendship at an impressive 16,240 ft.
- A multifaceted individual with passions spanning trekking, mountaineering, swimming, filmmaking, and fine arts.